Coursework 2 Final Submission Form and Report

HIJIN-BATTLE

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Unity version: 2022.3.62f3c1

Repository: https://github.com/Xiangtian-Gmae-program/Game_project

Repository folder: FINAL VERSION

2. Report

2.1 Design Choices

The game was designed as a focused samurai duel vertical slice rather than a large open-world action game. This choice kept the project realistic and made it possible to polish the main loop: start, choose mode/difficulty, fight, receive feedback, win or fail, retry, continue, or practise.

The visual style uses a night moon Japanese setting with dark UI panels and gold accents. This supports the intended mood of a disciplined sword duel and keeps the interface consistent with the scene atmosphere.

Practice mode was kept as a separate mode because it supports learning and testing. It records damage, hits, guard events, and evasion events, so the player can understand the combat system before entering the main game.

2.2 Technical Decisions

The technical approach was to extend the existing Unity project without a large architecture rewrite. This was important because the project already had working scenes, models, controllers, and animation references.

Technical Decision	Reason
Use MonoBehaviour scripts	Matches the existing Unity project style and keeps changes simple.
Use PlayerPrefs	Sufficient for lightweight difficulty, records, settings, and progression.
Use Resources loading	Allows BGM, SFX, fonts, story images, and enemy visuals to be found reliably in builds.
Separate enemy visual layer	Allows visual model replacement while keeping EnemyController as the gameplay brain.
Runtime UI controllers	Allows HUD, practice stats, story, pause, and result panels to be ensured in the right scenes.
Action locks	Prevents actions from interrupting each other and causing animation/movement conflicts.

2.3 Problems and Limitations

The largest problems were enemy animation stiffness, model compatibility, UI layout issues, and making the AI feel active without becoming unfair. Some imported models did not match the existing animation controller perfectly, so the enemy visual setup needed to be separated from the gameplay controller.

The game is currently a vertical slice. It has a playable loop and several polished systems, but it is not yet a full-length final game. More levels, more animations, richer sound effects, and additional balancing would be needed for a complete release.

2.4 Testing and What Changed Because of Testing

Test / Observation	Change Made
Practice panel did not appear	Fixed runtime UI/font creation and ensured PracticeStatsController was created correctly.
Enemy did not move or showed NavMesh warnings	Checked NavMeshAgent state and movement logic; adjusted enemy movement handling.
Enemy model disappeared after replacement	Restored visual resource loading and separated visual prefabs from gameplay logic.
Enemy defended too often	Reduced defense probability and increased attack pressure.
Hard mode still had passive gaps	Added close-range forced behaviour and made Hard mode more aggressive.
Guard recovery felt too fast	Slowed recovery and made guard bar changes visible.
Settings layout was cramped	Rebuilt the settings layout and added a themed opaque background.
Title/custom font caused layout issues	Repositioned title and selectively applied style font.

2.5 Reflection on Development from Concept to Final Version

The project developed from a combat prototype into a more complete vertical slice. Early work focused on understanding the existing scenes and transferring missing abilities into practice mode. Later work focused on main-game flow, enemy AI, UI feedback, audio, settings, story support, and submission preparation.

The strongest improvement was that the project now has a complete player journey: the player can start the game, practise controls, enter a main battle, understand the objective, fight with readable UI, pause, win or lose, and continue or retry.

2.6 Personal Contribution

My personal contribution includes design direction, combat system expansion, enemy AI tuning, UI design and implementation, practice statistics, scene flow, settings, audio integration, font/UI polish, testing, debugging, and documentation.

Contribution Area	Evidence
Gameplay	PlayerController.cs, EnemyController.cs, PlayerActionProfile.cs
UI	MainGameUIController.cs, PracticeStatsController.cs, UI/MenuPanelController.cs
Progression	MainGameProgress.cs, SceneFlowManager.cs
Audio	MainGameBgmController.cs, MainGameSfxController.cs, Resources/Audio
Visual polish	GameFontController.cs, SettingsPanelBackground.png, story images, UI layout changes
Documentation	README, CONTRIBUTION, checklist, manifest, portfolio/report documents

2.7 Use of Templates, Assets, Tutorials, or AI Support

External assets were used for models, environment, font, audio, and images. These were not submitted as unchanged work; they were integrated into a custom playable structure with original scene flow, scripts, UI, AI tuning, settings, testing, and documentation.

AI assistance was used for planning, debugging, code guidance, documentation organisation, and explanation. Outputs were reviewed, modified, and tested in Unity before being treated as part of the project. The final design direction, playtesting feedback, and project decisions were made by the project owner.

3. Evidence Against Game + Report Criteria

Assessment Area	Project Evidence
Playable and stable vertical slice	Title/menu, practice, maingame, HUD, win/fail, pause, settings, audio.
Clear player experience	Samurai night duel with readable controls and combat feedback.
Game systems	Input, game logic, collision/damage, animation, AI, UI, audio, settings, progression.
Programming decisions	Action locks, state-based enemy AI, Resources loading, PlayerPrefs, visual/gameplay separation.
Testing/debugging/improvement	Documented fixes for UI, NavMesh, model visibility, AI passivity, guard bar, layout, font.
Legal/ethical/accessibility/security	Asset credits needed, AI declaration included, local PlayerPrefs only, readable UI/controls.
Connection to design plan	Final result follows the vertical slice plan: practice, main battle, enemy AI, UI, audio, progression.

4. Game Demo / Presentation Support Plan

• Start from the title screen and explain the samurai night duel concept.

• Show the main menu, difficulty/records, and settings.

• Enter Practice Mode and show the statistics panel.

• Enter Main Game and explain HP, guard bar, controls panel, objective, and enemy target UI.

• Demonstrate attack, charged attack, defend, roll, jump, and pause.

- Explain enemy AI states: patrol, alert, chase, strafe, attack, defend, evade, stagger, return, dead.
- Show audio feedback: BGM, sword swing, and guard success.
- Answer honestly that more animation resources, SFX, and levels would be future improvements.